Neap

Trial Examination 2023

VCE Specialist Mathematics Units 3&4

Written Examination 1

Question and Answer Booklet

Reading time: 15 minutes
Writing time: 1 hour

Student's Name: _____

Teacher's Name: _____

Structure of booklet

Number of questions	*Number of questions to be answered*	*Number of marks*
10	10	40

Students are permitted to bring into the examination room: pens, pencils, highlighters, erasers, sharpeners and rulers.

Students are NOT permitted to bring into the examination room: any technology (calculators or software), notes of any kind, blank sheets of paper and/or correction fluid/tape.

Materials supplied

Question and answer booklet of 13 pages

Formula sheet

Working space is provided throughout the booklet.

Instructions

Write your **name** and your **teacher's name** in the space provided above on this page.

Unless otherwise indicated, the diagrams in this booklet are **not** drawn to scale.

All written responses must be in English.

At the end of the examination

You may keep the formula sheet.

Students are NOT permitted to bring mobile phones and/or any other unauthorised electronic devices into the examination room.

Students are advised that this is a trial examination only and cannot in any way guarantee the content or the format of the 2023 VCE Specialist Mathematics Units 3&4 Written Examination 1.

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Instructions

Answer **all** questions in the spaces provided.

Unless otherwise specified, an **exact** answer is required to a question.

In questions where more than one mark is available, appropriate working **must** be shown.

Unless otherwise indicated, diagrams in this booklet are **not** drawn to scale.

Take the **acceleration due to gravity** to have magnitude g ms^{-2}, where $g = 9.8$.

Question 1 (2 marks)

X and Y are independent random variables. The mean and the variance of X are 3 and 2, respectively; the mean and the variance of Y are 5 and 4, respectively.

Find the mean and variance of $X + 2X - 4Y$.

Question 2 (3 marks)

Given that $p, q \in Z$, consider the following statement.

If $p^2 + q^2 + 1$ is odd, then $p + q$ is even.

a. Write down the contrapositive of the statement. 1 mark

b. Prove that the contrapositive is true. 2 marks

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Question 3 (3 marks)

A spherical balloon is inflated such that its volume increases at a rate of 24 cm^3/s.

Determine the rate at which the balloon's surface area increases when the balloon's radius is 8 cm.

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Question 4 (3 marks)

Find the cube roots of –64. Express your answers in polar form using the principal values of the argument.

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Question 5 (3 marks)

Prove by mathematical induction that $1^3 + 2^3 + 3^3 + \cdots + n^3 = \frac{1}{4}n^2(n+1)^2$ for all $n \geq 1$, $n \in N$.

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Question 6 (6 marks)

A particle's position at any time t is given by $\underset{\sim}{r} = 2\cos(t)\underset{\sim}{i} - \sin(t)\underset{\sim}{j}$, where $t \geq 0$.

a. Find the Cartesian equation of the particle's path. 2 marks

b. Find the particle's speed when $t = \frac{\pi}{4}$. 2 marks

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c. Find the particle's maximum acceleration. 2 marks

Question 7 (4 marks)
Consider points $A(-2, 1, 0)$, $B(3, 4, -2)$ and $C(-5, -2, 1)$.

a. Use proof by contradiction to prove that points A, B and C are not collinear. 2 marks

b. Find the area of triangle ABC. 2 marks

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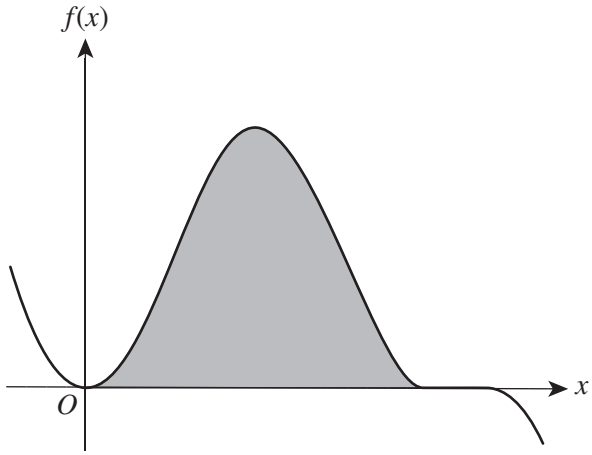
Question 8 (4 marks)

The graph of $f(x) = \sin^2(2x)\cos^3(2x)$ is shown below.



Find the shaded area.

Question 9 (4 marks)

Find $\int x^2 \sin(x)dx$.

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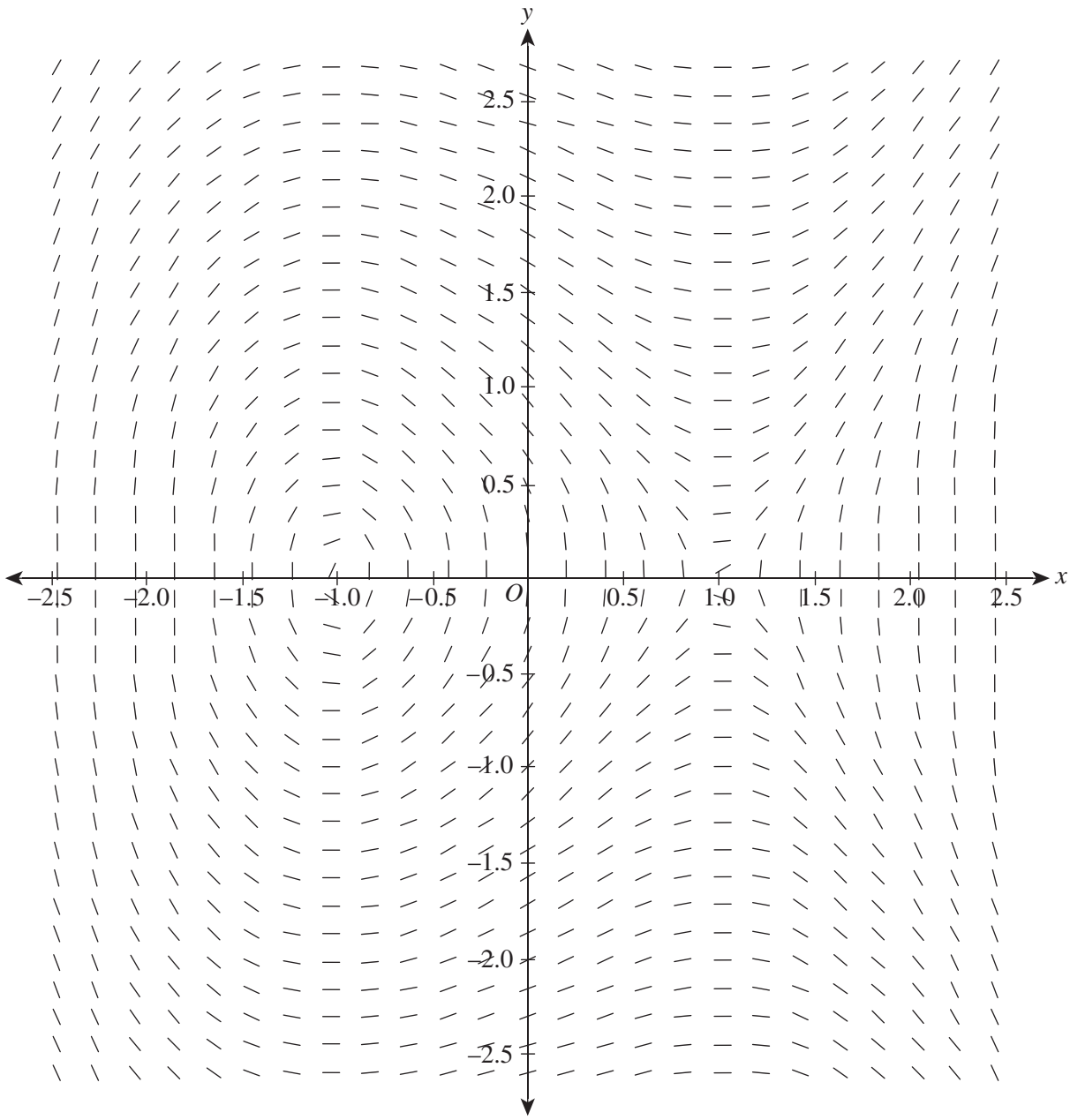
Question 10 (8 marks)

A direction field representing the differential equation $\frac{dy}{dx} = \frac{x^2 - 1}{y}$ is shown below.



a. On the direction field above, sketch the solution curve of the differential equation given that $y(0) = 2$. 2 marks

b. Solve the differential equation $\frac{dy}{dx} = \frac{x^2 - 1}{y}$ given that $y(0) = 2$. Express your answer in the form $ay^2 = bx^3 + cx + d$, where a, b, c and d are integers. 2 marks

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c. The solution curve of the differential equation $\frac{dy}{dx} = \frac{x^2 - 1}{y}$ has an inflection point such that $0 < x < 1$.

Show that the coordinates of the inflection point in the first quadrant satisfy the equation $y = \frac{1 - x^2}{\sqrt{2x}}$. 4 marks

END OF QUESTION AND ANSWER BOOKLET

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